MIL-C-14610G
17 October 1988
SUPERSEDING
MIL-C-14610F
21 April 1983

MILITARY SPECIFICATION

COVERALLS, EXPLOSIVES HANDLERS'

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers flame resistant cotton coveralls.

* 1.2 Classification. The coveralls shall be of one type in the following sizes (see 6.2):

Schedule of sizes

X-Small regular	Small short regular long	Medium short regular long	Large short regular long	X-Large short regular long

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 8415

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SPECIFICATIONS

FEDERAL

A-A-203	-	Paper, Kraft, Untreated
V-T-276	-	Thread, Cotton
DDD-L-20	-	Label: For Clothing, Equipage and Tentage, (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-F-21840	-	Fastener Tapes, Hook and Pile, Synthetic
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-C-43122	-	Cloth, Sateen, Cotton, Flame Retardant Treated
MIL-T-43548	-	Thread, Polyester Core: Cotton-, Rayon-, or Polyester-Covered

STANDARDS

FEDERAL

FED-STD-751	-	Stitches, Seams and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members for Containers and Pallets
MIL-STD-1668	-	Provisions for Evaluating Quality of Cloth Coveralls

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Copies should be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide samples. Guide samples of the end item, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this document may appear in the sample, in which case the document shall govern.

* 3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

* 3.3 Materials.

3.3.1 Basic material. The basic material shall be cotton sateen flame-retardant treated cloth conforming to class 1 of MIL-C-43122.

3.3.2 Thread. The thread shall be cotton conforming to type IA1, ticket No. 20, 4-ply, ticket No. 30, 3-ply, and ticket No. 50, 3-ply, of V-T-276, and shall be used in accordance with table II operations. Polyester cotton-covered thread, white or natural, conforming to MIL-T-43548 may be substituted for the cotton thread. Ticket nos. 50 and 30 shall be utilized where equivalent sizes of cotton are specified, except that ticket No. 70-2 may be used for all overedge operations, and 30-3 in lieu of ticket No. 20-4 cotton.

3.3.3 Fastener tapes. The fastener tapes for the front closure, side entries, and sleeve adjustments shall be 1-inch wide hook tape and 1-inch wide pile tape conforming to type II, class 1 of MIL-F-21840. The color shall be white 3054.

3.3.4 Labels. Each coverall shall have a size label, identification label, and instruction label, or a combination identification and instruction label, conforming to DDD-L-20. The fastness to laundering requirements of DDD-L-20 shall apply.

3.3.4.1 Size label. The size label shall conform to type VI, class 2.

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3.3.4.2 Identification label. The identification label shall conform to type VI, class 1.

3.3.4.3 Instruction label. The instruction label shall conform to type VI, class 3. The body of the instruction label shall be as follows:

Wear as outer garment.
Keep front closed to assure body protection.
Keep sleeves adjusted.
Keep pocket clean and free of dust accumulation.
Wash at low temperature. Dry thoroughly.
Warning: Upon removal of garment, brush off accumulated foreign matter thoroughly.

3.3.4.4 Combination label (when applicable). The identification label (type VI, class 1) and instruction label (type VI, class 3) may be combined into one label. The combination label shall conform to the requirements of type VI, class 15 label.

3.4 Design. The coveralls have a shirt-type collar, long sleeves with adjustment tapes, a lattice hip pocket on right side, and side entries. The front and side entries are closed with fastener tapes.

3.5 Figure. Figure 1 is furnished for information only. To the extent of any inconsistencies between the written document and the figure, the document shall govern.

3.6 Patterns. Standard patterns, to be used to cut the working patterns, will be furnished by the Government. The working patterns shall be identical to the Government patterns which shall not be altered in any way, except as indicated in footnotes to table I. The patterns provide a seam allowance of 1/2 inch for double-lapped, double-stitched seams and 3/8 inch for all other seams.

3.6.1 List of pattern parts. The coveralls shall be cut from the materials specified in accordance with the number of parts indicated in table I.

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TABLE I. List of pattern parts

Component	Nomenclature	Cut parts
Basic material	Top front (waist)	2
	Top back (waist) 1/	1
	Trouser front	2
	Trouser front extension (side entry)	2
	Trouser back	2
	Right front facing	1
	Sleeve	2
	Collar 2/	2

1/ Back may be cut in two pieces. When cut in two pieces, a seam allowance of 1/2 inch shall be added for a double-lapped, double-stitched seam.

2/ The collar may be cut in one piece by eliminating the bottom edge seam allowance and abutting the bottom edge of two collar parts. The collar width from neck edge to bottom edge at center of collar shall finish not less than 4-1/4 inches in the finished coveralls.

3.7 Construction.

3.7.1 Stitches, seams, and stitching. Stitch, seam, and stitching types specified in table II shall conform to FED-STD-751. When two or more methods of seaming or stitch type are given for the same operation, the method and type are optional. Where stitch type 401 is used, the looper (underthread) shall be on the inside of the coveralls. The gage of stitching for double-lapped, double-stitched seams shall be 1/4 to 5/16 inch. All seams shall start and finish evenly. The stitching on the fastener tapes shall be not less than 1/16 inch from each edge.

3.7.1.1 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in table II.

3.7.1.2 Seam type, Seam type shall be as specified in table II with seam allowances maintained and seams properly sewn so that no raw edges, run-offs, twists, pleats, puckers, or open seams shall result.

3.7.1.3 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched 1/4 inch minimum except when caught in other stitching or turned under in a hem. Thread tension shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the materials sewn.

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3.7.1.3.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching. 1/

b. Thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area, continue over the defective area, and continue a minimum of 1/2 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.7.1.4 Type 401, 406, 502, 503, 504, 515, 516, 517, 518, and 519 stitching. Thread tension shall be maintained so that there will be no loose stitching. All repairs shall be in accordance with 3.7.1.3.1a or 3.7.1.3.1b. Thread tension shall be maintained so that there will be no loose or excessively tight stitching resulting in puckering of the materials sewn.

3.7.2 Bartacking. Unless otherwise specified, bartacks shall be as follows:

		Tolerance		
Length	Width	Length	Width	Stitches per bartack
1/2 inch	1/8 inch	± 1/16 inch	± 1/32 inch	28
1 inch	1/8 inch	± 1/16 inch	± 1/32 inch	42

Bartacking shall be free from thread breaks and loose or tight stitching.

3.7.3 Thread ends. All thread ends shall be trimmed to 1/4 inch maximum length.

* 3.8 Manufacturing operations requirements. The coveralls shall be manufactured in accordance with operation requirements specified in table II. The contractor is not required to follow the exact sequence of operations listed.

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TABLE II.

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. a. Cut the coverall in accordance with patterns which show directional lines, marks, and notches for proper assembly. Directional lines shall be placed in the warp direction of the material. The directional lines on the fronts and backs of the trouser parts may vary from the warp direction by not more than 2-1/2 inches. Measurements shall be taken from the top and bottom of the directional line on the pattern to the selvage edge of the fabric, and the difference between the two measurements shall not exceed 2-1/2 inches. The collar may be cut in either the warp or filling direction. The filling-effect side of the material shall finish on the outside of the garment. b. Cut all component parts from the same piece of material, except the hanger, lattice pocket stripping, undercollar (when applicable), and right front facing, which may be cut from ends. c. Cut the stripping for the hanger loop of sufficient width to finish 3/8 to 1/2 inch wide (see operation 4). d. Cut the stripping for the lattice pocket of sufficient width to finish 5/8 to 1 inch wide (see operation 4).						

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
2.	Replacement of damaged parts. a. Care shall be exercised during spreading, cutting, and manufacturing operations to ensure that material defects and damages, as classified in MIL-STD-1668, are excluded and replaced with non-defective and properly matched material.						
3.	Marking. a. Mark, ticket, or bundle component parts of the coverall, except hanger loop and lattice pocket stripping, to ensure a correct uniform size and proper assembly throughout the coverall. b. Any method of marking may be used except: 1. Metal fastening devices. 2. Sew-on tickets. 3. Adhesive type tickets which leave traces of adhesive on the material after removal of the ticket.						
4.	Make hanger loop and lattice pocket stripping (see figure 1). a. Turn edges of the hanger stripping (see 1.c) and lattice pocket stripping (see 1.d) to the center and stitch the abutted edges with a row of stitches (1/4 to 5/16 inch gage) extending across the covering edges of the material. - or -	406	EFh-1	10-14	50-3	50-3	

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
4.	Make hanger loop and lattice pocket stripping (see figure 1). (cont'd)						
	b. Fold stripping with the raw edges turned in and stitch 1/16 to 1/8 inch from each edge.	301	EFp-2	10-14	30-3	30-3	
	- or -						
	c. Fold stripping with raw edges turned in and stitch 1/16 to 1/8 inch from turned in edge.	301	EFp-1	10-14	30-3	30-3	
	- or -						
	d. Turn in the edges of two pieces of stripping and stitch together with a row of stitching 1/16 to 1/8 inch from each edge.	301	EFn-2	10-14	30-3	30-3	
	e. Turn a strip of material, folding one edge inwardly, and join with two rows of stitching 1/16 to 1/8 inch from each edge. (If option to use safety stitching is exercised, overedging shall cover exposed raw edge of material. If 301 option is used, raw edge of material shall be selvaged or overedged.)	301 or 517 or 518	EFz-2 EFz-2 EFz-2	10-14 10-14 10-14	30-3 50-3 50-3	30-3 50-3 50-3	
5.	Overedging.						
	a. Overedge the following edges if not selvaged:	502 or	SSa-1	6-10	50-3	50-3	
	1. Front and bottom edge of front extension piece.	503 or	SSa-1	6-10	50-3	50-3	
	2. Edge of side entry on trouser fronts.	504	SSa-1	6-10	50-3	50-3	
	3. Front edge of front waist.						
	4. Front edge of left trouser.						
	5. Back edge of right front facing.						

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
6.	Make sleeves.						
	a. Hem sleeves with a raw edge turned in. The hem shall finish 1 ± 1/4 inch wide. This operation may be performed after seaming underarms.	301 or 401	EFb-1 EFb-1	10-14 10-14	30-3 30-3	30-3 50-3	
	b. Stitch a 2-1/2 inches ± 1/4-inch pile tape on front of sleeve and a 2-1/2 inches ± 1/4-inch hook tape on back of sleeve, positioned 2-1/2 inches ± 1/8 inch from the bottom of hemmed sleeve and so that one end will be caught in the underarm seam (see operation 18).	301 301	SSau-1 SSau-1	10-14 10-14	30-3 or 20-4	30-3 30-3	
7.	Make collar.						
	a. Stitch collar ends (one-piece collar) or top collar and under collar together (two-piece collar). Trim points, work out points and edges, turn collar, and edge stitch 3/16 to 1/4 inch from edge. The edge stitching shall not extend to the ends of collar to allow for collar setting. - or -	301 or 401 and 301 301	SSe-2(a) SSe-2(a) SSe-2(b) or OSf-1	10-14 10-14 10-14 10-14	30-3 30-3 30-3 30-3	30-3 50-3 30-3 30-3	
	b. When one-piece collar is used, fold collar in half lengthwise, turn in raw edges at ends, and stitch with two rows of stitching, 3/16 to 1/4 inch gage, with the first row 1/16 to 1/8 inch from edge. Continue stitching across bottom edge of collar. The stitching shall not extend to ends of collar to allow for setting. A single row of stitching may be used across bottom edge of collar, 3/16 to 1/4 inch from folded edge. This operation may be performed when setting collar (operation 16).	301 301 301 301	SSc-2 and OSf-2 or SSc-2 and OSf-1	10-14 10-14 10-14 10-14	30-3 30-3 30-3 30-3	30-3 30-3 30-3 30-3	

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
8.	Make lattice pocket (see figure 1).						
	a. Cut the lattice pocket stripping (see operation 4) into five pieces of equal length, 8 to 8-1/4 inches long. The pocket consists of three vertical and two horizontal strips.						
	b. Position one horizontal strip across and even with the top edges of three vertical strips. The ends of the horizontal strip shall extend 1/2 ± 1/8 inch beyond the ends of the vertical strips and the vertical strips shall be equidistant apart. Place the second horizontal strip 4 inches down from the top of the top strip (1/2 inch off center tolerance). Bartack the horizontal strips to the vertical strips at each crossing with 1/2-inch horizontal bartacks along the upper edge of the horizontal strips (6 bartacks). The horizontal strips may be stitched to the vertical strips prior to bartacking.	Bartack		28 per bartack	30-3 or 50-3	30-3 or 50-3	
9.	Make trouser backs.						
	a. Position pocket on right back of trouser as indicated by marks on pattern and bartack the lower and side ends of each latticed strip to the trousers (7 bartacks). The bartacks shall be 1/2 inch long. The pocket shall finish 7 to 7-1/4 inches wide and 7-1/2 to 7-3/4 inches long.	Bartack		28 per bartack	30-3 or 50-3	30-3 or 50-3	
	b. Join seat seam with a double-lapped, double-stitched seam. An additional line of stitching through the center of the double-lapped, double-stitched seam shall be acceptable.	401	LSc-2	10-14	30-3	50-3	

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
9.	Make trouser backs. (cont'd)						
	- or -						
	c. Safety-stitch seat seam.	515 or 516 or 517 or	SSa-2	10-14	30-3 (chain stitching) and 50-3 (overedging)	50-3	
		519	SSa-2	10-14	20-4 (all thread)	20-4	
10.	Make side entries on front trousers.						
	a. Stitch the hook part of the fastener tape to the front extension piece with the front edge of the tape not more than 1 inch nor less than 7/8 inch from the front edge (overedged or selvage) of the front extension piece.	301	SSau-2	10-14	30-3	30-3	
	b. Turn in the side entry on each trouser front forming a facing. Stitch the pile part of the fastener tape on facing through trouser front. The back edge of the tape shall be 1/4 to 5/16 inch from the back edge of facing.	301	SSau-2	10-14	30-3 or	30-3	
		301	SSau-2	10-14	20-4	30-3	
	c. Fasten the hook part of tape (front extension) to the pile part (trouser front) and stitch across top of entry catching the ends of the fastener tapes in the stitching and overlapping the ends of the stitching. On the finished coverall, the stitching shall be 3/4 $\pm$ 1/4 inch from the botton edge of the double-lapped waist seam.	301	SSau-2	10-14	30-3 or	30-3	
		301	SSau-2	10-14	20-4	30-3	

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					NEEDLE	BOBBIN/ LOOPER	COVER
10.	Make side entries on front trousers. (cont'd)						
	d. Stitch across bottom of entry catching the ends of the fastener tapes in the stitching. With the raw edge turned in, continue the stitching diagonally to side edge of the trouser, across bottom, catch bottom of extension piece in the stitching and stitch diagonally back to the bottom of the side entry (forming a trapezoid pattern).	301 301	LSd-1 and SSv-1	10-14 10-14	30-3 30-3	30-3 30-3	
11.	Join back waist and attach labels.						
	a. If back waist is cut in two pieces (see 3.6.1), join back sections together with a double-lapped, double-stitched seam. The seam may be lapped in either direction. An additional row of stitching through the center of the double-lapped seam shall be acceptable.	301 or 401	LSc-2 LSc-2	10-14 10-14	30-3 30-3	30-3 50-3	
	b. Position the size label on the center back 1 ± 1/4 inch from the neck raw edge.	301	LSbj-1	10-14	30-3	30-3	
	c. Center the identification label not more than 3/4 inch below the size label.	301	LSbj-1	10-14	30-3	30-3	
	d. Center the instruction label (or combination identification and instruction label) not more than 3/4 inch below the size label or identification label, as applicable.	301	LSbj-1	10-14	30-3	30-3	
	NOTE: The stitching shall not be through the printing. A 1-inch off center tolerance shall be allowed. The labels shall be stitched on all four sides. The stitching of one label may catch the top or bottom of another label.						

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
12.	Join trousers to waist.						
	a. Join back waist to trouser back, left waist to left trouser front, and right waist to right trouser front with a double-lapped, double-stitched seam. An additional row of stitching through the center of the double-lapped, double-stitched seam shall be acceptable. The waist shall lap the trousers on outside.	301 or 401	LSc-2 LSc-2	10-14 10-14	30-3 30-3	30-3 30-3	
13.	Make fronts.						
	a. Stitch right facing to front from collar notch, across step and down front. Trim point, slash facing at notch, turn, and edge stitch 1/8 to 1/4 inch from edge. The step on the finished coverall shall be 1-1/2 inches ± 1/8 inch wide.	301 or 401 or 301	SSe-2(a) SSe-2(a) SSe-2(b)	10-14 10-14 10-14	30-3 30-3 30-3	30-3 50-3 30-3	
	b. Position hook fastener tape on outside of coverall with the front edge 5/8 ± 1/8 inch from edge of front and stitch to front through facing.	301	SSau-2	10-14	30-3	30-3	
	c. Turn in left facing, place pile fastener tape on facing with the front edge 1/8 inch from edge of front, and stitch to facing through front.	301	SSau-2	10-14	30-3	30-3	
14.	Join crotch seam.						
	a. Turn raw edge of crotch seam allowance to the outside and stitch 1/16 to 1/4 inch from edge.	301 or 401	LSdc-3(a) LSdc-3(a)	10-14 10-14	30-3 30-3	30-3 30-3	
	b. Turn in raw edge of left front crotch seam allowance and stitch left front to right front with three rows of stitching 1/8 inch apart.	301 or 401	LSdc-3(b) LSdc-3(b)	10-14 10-14	30-3 30-3	30-3 50-3	

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					NEEDLE	BOBBIN/ LOOPER	COVER
14.	Join crotch seam. (cont'd)						
	c. Box stitch top of crotch seam enclosing the ends of the fastener tapes within the box. The box pattern shall be 1 ± 1/8 inch wide and 1/2 ± 1/8 inch deep.	301		10-14	30-3	30-3	
	- or -						
	d. Bartack across top of crotch seam, securing both rows of stitching with one 1-inch or two 1/2-inch bartacks. The bartack(s) shall be positioned 1/4 inch above the bottom finished edges of the fastener tape.	Bartack		42 per bartack	30-3 or 50-3	30-3 or 50-3	
		Bartack		28 per bartack	30-3 or 50-3	30-3 or 50-3	
15.	Join shoulder seams.						
	a. Join shoulders with double-lapped, double-stitched seams with backs overlapping the fronts. An additional line of stitching through the double-lapped, double-stitched seam shall be acceptable.	301 or	LSc-2	10-14	30-3	30-3	
		401	LSc-2	10-14	30-3	50-3	
	- or -						
	b. Safety-stitch shoulder seams.	515 or 516 or 517 or	SSa-2	10-14	30-3 (chain stitching) and 30-3 (overedging)	50-3	
		519	SSa-2	10-14	20-4 (all thread)	20-4	

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					NEEDLE	BOBBIN/ LOOPER	COVER
16.	Set collar.						
	a. Stitch undercollar to coveralls from collar notch on right front to end of left front.	301 or 401	SSa-1 SSa-1	10-14 10-14	30-3 30-3	30-3 50-3	
	b. Turn in the lower edge of top collar and stitch through all plies 1/16 to 1/8 inch from edge, catching ends of the hanger loop in the stitching at the center back (3/4 inch off center tolerance). Continue stitching at each end to complete the edge stitching of the collar (see operation 7). If the back is two-piece, the hanger loop shall be positioned to right or left of the center back seam. The hanger loop shall measure 1-1/2 inches ± 1/4 inch long (folded) when attached.	301	LSb-1	10-14	30-3	30-3	
17.	Set sleeves.						
	a. Set sleeves with a double-lapped, double-stitched seam. An additional line of stitching through the double-lapped, double-stitched seam shall be acceptable. The fronts and back shall lap the sleeves on outside.	301 or 401	LSc-2 LSc-2 or	10-14 10-14	30-3 30-3	30-3 50-3	
	- or -						
	b. Set sleeve with safety-stitch.	515 or 516 or 517 or	SSa-2	10-14	30-3 (chain stitching) and 50-3 (overedging)	50-3 50-3	
		519	SSa-2	10-14	20-4 (all thread)	20-4	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
18.	Join side seams.						
	a. Join side seam (waist and trouser) and underarm seams with double-lapped, double-stitched seams securely catching the ends of the sleeve fastener tapes in the stitching. An additional line of stitching through the center of the double-lapped, double-stitched seam shall be acceptable.	301 or 401	LSc-2 LSc-2	10-14 10-14	30-3 30-3	30-3 50-3	
	- or -						
	b. Safety-stitch side and underarm seams from bottom of sleeves to bottom of legs.	515 or 516 or 517 or	SSa-2	10-14	30-3 (chain stitching) and 50-3 (overedging)	50-3 and 50-3	
	NOTE: The seams at underarms and waist shall not be staggered more than 3/8 inch, measured from center of seam to center of seam.	519	SSa-2	10-14	20-4 (all thread)	20-4	
19.	Join inseams.						
	a. Join inseams with a double-lapped, double-stitched seam. The seams at crotch shall not be staggered more than 1/2 inch (measured from center of seam to center of seam). An additional line of stitching through the center of the double-lapped, double-stitched seam shall be acceptable.	301 or 401	LSc-2 LSc-2	10-14 10-14	30-3 30-3	30-3 50-3	
20.	Hem trousers.						
	a. Hem bottom of trousers with the raw edge turned in and the stitching 1/16 to 1/8 inch from edge. The hem shall finish 1 ± 1/4 inch wide. The bottoms may be hemmed prior to joining inseams (see operation 19).	301	EFb-1	10-14	30-3	30-3	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
21.	Bartacks (sleeves and trousers bottoms).						
	a. Applicable when option to hem sleeves before underarm seaming or to hem trouser legs before inseaming. Place a horizontal 1/2-inch bartack across each joining seam.	Bartack		28 per bartack	30-3 or 50-3	30-3 or 50-3	
22.	Clean coveralls.						
	a. Trim all ends of stitching and remove loose threads.						
	b. Remove all spots, stains, and size marking.						

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* 3.9 Measurements. The finished coverall shall conform to the finished measurements shown in table III. The tolerance for all measurements shall be ± 1/2 inch.

TABLE III. Finished coverall measurements (inches)

	X-small	Small			Medium			Large			X-large		
	reg.	short	reg.	long	short	reg.	long	short	reg.	long	short	reg.	long
1/2 chest 1/	20-1/2	22-1/2	22-1/2	22-1/2	24-1/2	24-1/2	24-1/2	26-1/2	26-1/2	26-1/2	28-1/2	28-1/2	28-1/2
1/2 waist 2/	17-1/2	19-1/2	19-1/2	19-1/2	21-1/2	21-1/2	21-1/2	23-1/2	23-1/2	23-1/2	25-1/2	25-1/2	25-1/2
Sleeve inseam 3/	23	22	23	24	22	23	24	22	23	24	22	23	24
Leg inseam 4/	32	30	32	34	30	32	34	30	32	34	30	32	34
Overall length 5/	64-1/2	64-1/2	65-1/2	66-1/2	65-1/2	66-1/2	67-1/2	66-1/2	67-1/2	68-1/2	67-1/2	68-1/2	69-1/2

1/ With coverall closed, measure across front at base of armhole from folded edge to folded edge.

2/ With coverall closed, measure across waist from folded edge to folded edge.

3/ Measure sleeve inseam from base of armhole to bottom of sleeve.

4/ Measure leg inseam from crotch seam to bottom of leg.

5/ Measure from base of neck along center to bottom of leg.

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3.10 Workmanship. The finished coverall shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

* 4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

* 4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in MIL-STD-1668. The presence of any defect shall be cause for rejection of the first article.

* 4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2 Point count inspection. Sampling and inspection provisions for end item examination, dimensional examination, and packaging inspection shall be performed in accordance with MIL-STD-1668.

* 4.4.3 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each pair of fully closed coveralls shall be folded to approximately 23 by 14-1/2 inches in such a manner that the size label is clearly visible to permit ready identification.

* 5.1.2 Commerical. Coveralls shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

* 5.2.1 Level A packing. Fifteen coveralls of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container, conforming to style RSC-L, grade V2s, of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Level A

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unit packs shall be packed flat fifteen in depth, with every other one reversed within the shipping container. Inside dimenions of each container shall be approximately 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to A-A-203. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.2. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Fifteen coveralls of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container, conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275, of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Level A unit packs shall be packed flat, 15 in depth, with every other one reversed within a shipping container. Inside dimensions of each container shall be approximately 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to A-A-203. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.2.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.2.

* 5.2.3 Commercial packing. Coveralls, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Palletization. When specified (see 6.2), coveralls, packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be number 3 in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

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* 5.4 Marking. In addition to any special marking required by the contract or purchase order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The coveralls are intended for wear by personnel engaged in operations involving the handling of ammunition or explosives.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Sizes required (see 1.2).
c. When a first article is required (see 3.2, 4.3, and 6.4).
d. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
e. Type and class of unit load required (see 5.2.1).
f. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
g. When palletization is required (see 5.3).

6.3 Guide samples. For access to sample of end item and shade samples, address the procuring activity issuing the invitation for bids.

* 6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

* 6.5 Subject term (key word) listing.

Clothing
Coveralls
Explosives handlers'

6.6 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.

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Air Force - 99

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DLA - CT

User activities:

Navy - OS, CG
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Preparing activity:

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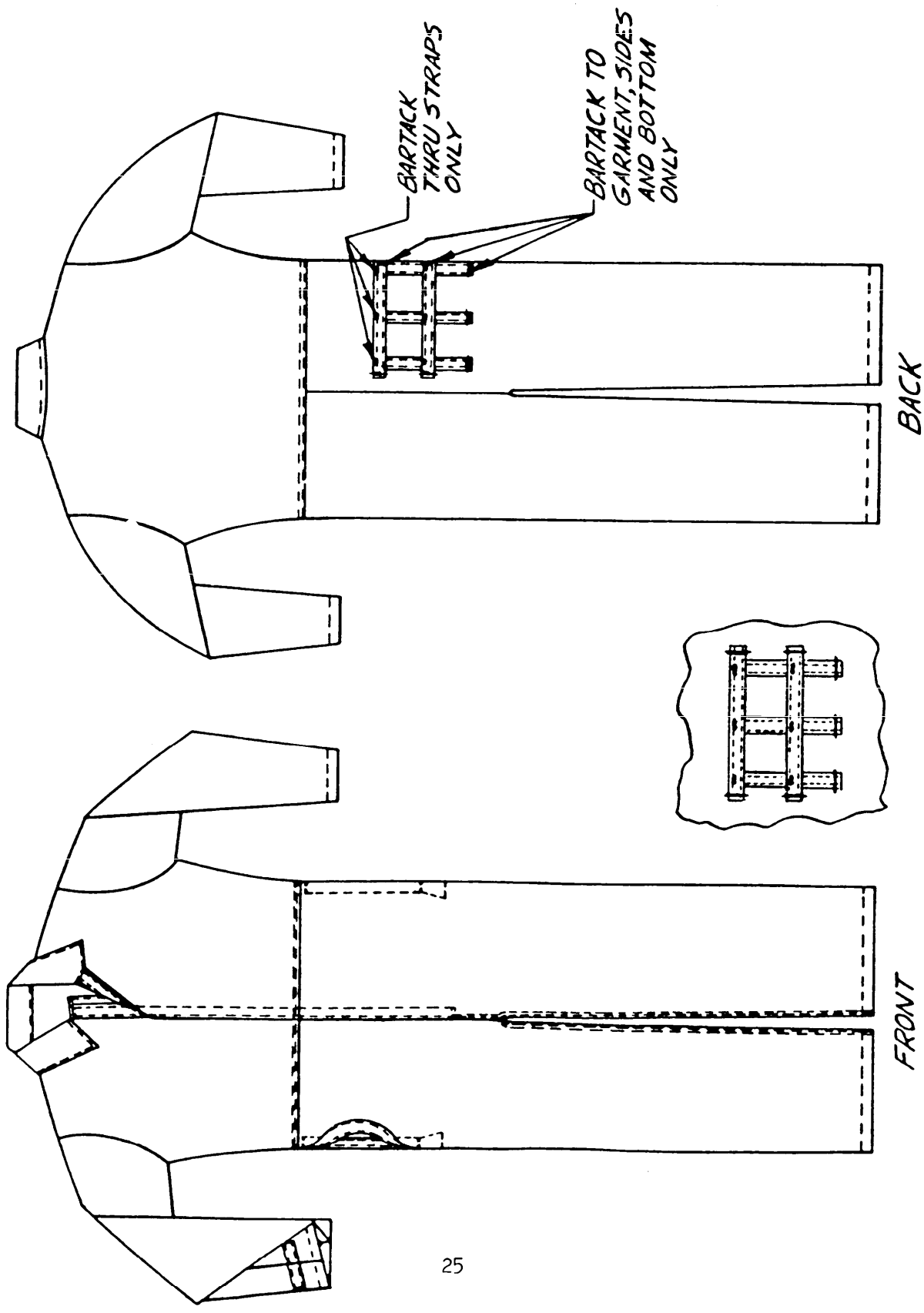


FIGURE 1. Coveralls, Explosives Handlers'.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-14610G	Coveralls, Explosives Handlers'

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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